

(bars: lengths by R , gain by a_{nom} ; hats on estimates; $e_\theta = \theta - \hat{\theta}$; posterior quantities unless marked $-$; $\bar{w}_d[k]$ commanded height, $\Delta \bar{w}_d[k] = \bar{w}_d[k+1] - \bar{w}_d[k]$; $\bar{a}' := \bar{a}'_{\text{true}}(\bar{w}_d[k] - \delta \hat{w}_3[k])$, $\bar{a}'' := \bar{a}''_{\text{true}}(\bar{w}_d[k] - \delta \hat{w}_3[k])$, both exogenous; $\Delta \bar{w}[k] := \bar{w}[k+1] - \bar{w}[k]$; $\Delta \hat{w}[k] := \Delta \bar{w}_d[k] + (1 - \lambda_c)[\delta \hat{w}_3[k] + y_1[k] - \delta \hat{w}_1[k]]$; $e_{\Delta \bar{w}} := \Delta \bar{w} - \Delta \hat{w}$; $\mathbb{E}[\cdot]$ mean over noise realizations; $P_{ij} = \mathbb{E}[e_i e_j]$; $d = 2$)

1 True dynamics

$$\begin{aligned}
\bar{w}[k] &= \bar{w}_d[k] - \delta \bar{w}_3[k] \\
\delta \bar{w}_3[k+1] &= \lambda_c \delta \bar{w}_3[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \varepsilon_w[k] \\
\Delta \bar{w}[k] &= \Delta \bar{w}_d[k] + (1 - \lambda_c) \delta \bar{w}_3[k] + F_{dw}[k] e_{\bar{a}_w}[k] + \varepsilon_w[k] \\
\bar{a}_w[k+1] - \bar{a}_w[k] &= \bar{a}'_{\text{true}}(\bar{w}[k]) \Delta \bar{w}[k] + \frac{1}{2} \bar{a}''_{\text{true}}(\bar{w}[k]) \Delta \bar{w}[k]^2 + \mathcal{O}(\Delta \bar{w}^3) \\
\bar{a}'_{\text{true}}(\bar{w}[k]) &= \bar{a}'_{\text{true}}(\bar{w}_d[k] - \delta \hat{w}_3[k] - e_{\delta \bar{w}_3}[k]) = \bar{a}' - \bar{a}'' e_{\delta \bar{w}_3}[k] + \mathcal{O}(e^2) \\
\bar{a}''_{\text{true}}(\bar{w}[k]) &= \bar{a}'' + \mathcal{O}(e)
\end{aligned}$$

$$\bar{a}_w[k+1] - \bar{a}_w[k] = (\bar{a}' - \bar{a}'' e_{\delta \bar{w}_3}[k]) \Delta \bar{w}[k] + \frac{1}{2} \bar{a}'' \Delta \bar{w}[k]^2$$

$$\begin{aligned}
F_{dw}[k] &= \bar{f}_{dw}[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{f}_{dw}[k-i] \\
\varepsilon_w[k] &= \bar{a}_w[k] \bar{f}_T[k] + (1 - \lambda_c) \sum_{i=1}^d \bar{a}_w[k-i] \bar{f}_T[k-i] + (1 - \lambda_c) n_w[k] \\
y_1[k] &= \delta \bar{w}_1[k] + n_w[k]
\end{aligned}$$

$$\begin{aligned}
\mathbb{E}[\varepsilon_w[k] n_w[k]] &= (1 - \lambda_c) R_1 \\
\tilde{\varepsilon}_w[k] &:= \varepsilon_w[k] - (1 - \lambda_c) n_w[k], \quad \mathbb{E}[\tilde{\varepsilon}_w n_w] = 0, \quad \mathbb{E}[\tilde{\varepsilon}_w^2] = Q_{33} - (1 - \lambda_c)^2 R_1 =: \tilde{Q}_{33} \\
\Delta \bar{w}[k] &= \Delta \bar{w}_d[k] + (1 - \lambda_c) [\delta \bar{w}_3[k] + n_w[k]] + F_{dw}[k] e_{\bar{a}_w}[k] + \tilde{\varepsilon}_w[k]
\end{aligned}$$

2 Estimator

$$\begin{aligned}
\delta \hat{w}_3[k+1] &= \lambda_c \delta \hat{w}_3[k] - (1 - \lambda_c) [y_1[k] - \delta \hat{w}_1[k]] \\
\Delta \hat{w}[k] &= \Delta \bar{w}_d[k] + (1 - \lambda_c) [\delta \hat{w}_3[k] + y_1[k] - \delta \hat{w}_1[k]] \\
\hat{a}_w[k+1] - \hat{a}_w[k] &= \bar{a}' \Delta \hat{w}[k] + \frac{1}{2} \bar{a}'' \Delta \hat{w}[k]^2
\end{aligned}$$

3 Error increment

$$y_1[k] - \delta \hat{w}_1[k] = e_{\delta \bar{w}_1}[k] + n_w[k]$$

$$e_{\Delta \bar{w}}[k] = \Delta \bar{w}[k] - \Delta \hat{w}[k] = (1 - \lambda_c)[e_{\delta \bar{w}_3}[k] - e_{\delta \bar{w}_1}[k]] + F_{dw}[k] e_{\bar{a}_w}[k] + \tilde{\varepsilon}_w[k]$$

$$e_{\bar{a}_w}[k+1] - e_{\bar{a}_w}[k] = (\bar{a}_w[k+1] - \bar{a}_w[k]) - (\hat{a}_w[k+1] - \hat{a}_w[k])$$

$$= (\bar{a}' - \bar{a}'' e_{\delta \bar{w}_3})(\Delta \hat{w} + e_{\Delta \bar{w}}) + \frac{1}{2} \bar{a}'' (\Delta \hat{w} + e_{\Delta \bar{w}})^2 - \bar{a}' \Delta \hat{w} - \frac{1}{2} \bar{a}'' \Delta \hat{w}^2$$

$$= \bar{a}' e_{\Delta \bar{w}} - \bar{a}'' e_{\delta \bar{w}_3} \Delta \hat{w} - \bar{a}'' e_{\delta \bar{w}_3} e_{\Delta \bar{w}} + \bar{a}'' \Delta \hat{w} e_{\Delta \bar{w}} + \frac{1}{2} \bar{a}'' e_{\Delta \bar{w}}^2 + \mathcal{O}(3)$$

4 Mean

$$\mathbb{E}[e_{\delta \bar{w}_1}] = \mathbb{E}[e_{\delta \bar{w}_3}] = \mathbb{E}[e_{\bar{a}_w}] = \mathbb{E}[\tilde{\varepsilon}_w] = 0$$

$$\mathbb{E}[e_{\delta \bar{w}_3}^2] = P_{33}, \quad \mathbb{E}[e_{\delta \bar{w}_1}^2] = P_{11}, \quad \mathbb{E}[e_{\delta \bar{w}_1} e_{\delta \bar{w}_3}] = P_{13}$$

$$\mathbb{E}[e_{\delta \bar{w}_3} e_{\bar{a}_w}] = P_{34}, \quad \mathbb{E}[e_{\delta \bar{w}_1} e_{\bar{a}_w}] = P_{14}, \quad \mathbb{E}[e_{\bar{a}_w}^2] = P_{44}$$

$$\mathbb{E}[\tilde{\varepsilon}_w e_{\delta \bar{w}_1}] = \mathbb{E}[\tilde{\varepsilon}_w e_{\delta \bar{w}_3}] = \mathbb{E}[\tilde{\varepsilon}_w e_{\bar{a}_w}] = \mathbb{E}[\tilde{\varepsilon}_w n_w] = 0$$

$$\mathbb{E}[e_j \delta \hat{w}_i] = \mathbb{E}[e_j y_1] = 0 \quad (\text{posterior error} \perp \text{data})$$

$$\mathbb{E}[\bar{a}' e_{\Delta \bar{w}}] = 0$$

$$\mathbb{E}[-\bar{a}'' e_{\delta \bar{w}_3} \Delta \hat{w}] = 0$$

$$\mathbb{E}[-\bar{a}'' e_{\delta \bar{w}_3} e_{\Delta \bar{w}}] = -\bar{a}'' [(1 - \lambda_c)(P_{33} - P_{13}) + F_{dw} P_{34}]$$

$$\mathbb{E}[\bar{a}'' \Delta \hat{w} e_{\Delta \bar{w}}] = 0$$

$$\mathbb{E}[\frac{1}{2} \bar{a}'' e_{\Delta \bar{w}}^2] = \frac{1}{2} \bar{a}'' [(1 - \lambda_c)^2 (P_{33} + P_{11} - 2P_{13}) + 2(1 - \lambda_c) F_{dw} (P_{34} - P_{14})$$

$$+ F_{dw}^2 P_{44} + \tilde{Q}_{33}]$$

$$\mathbb{E}[e_{\bar{a}_w}[k+1]] - \mathbb{E}[e_{\bar{a}_w}[k]] = \bar{a}'' \left\{ -(1 - \lambda_c)(P_{33} - P_{13}) - F_{dw} P_{34} \right.$$

$$+ \frac{1}{2} (1 - \lambda_c)^2 (P_{33} + P_{11} - 2P_{13}) + (1 - \lambda_c) F_{dw} (P_{34} - P_{14})$$

$$\left. + \frac{1}{2} F_{dw}^2 P_{44} + \frac{1}{2} \tilde{Q}_{33} \right\}$$

(first, second and fourth lines: $\bar{a}'$ and $\Delta \hat{w}$ are functions of the data, $e_{\Delta \bar{w}}$ and $e_{\delta \bar{w}_3}$ are posterior errors; without the $y_1 - \delta \hat{w}_1$ input these four means are $\mp \bar{a}''(1 - \lambda_c) \ell_{31} R_1$ and cancel only in pairs, 0902_formC_aptrue_4state.tex, Error dynamics)

5 Estimator with the mean term

$$\begin{aligned}\hat{a}_w[k+1] &= \hat{a}_w[k] + \bar{a}' \Delta \hat{w}[k] + \frac{1}{2} \bar{a}'' \Delta \hat{w}[k]^2 \\ &\quad + \bar{a}'' \left\{ -(1 - \lambda_c)(P_{33}[k] - P_{13}[k]) - F_{dw}[k] P_{34}[k] \right. \\ &\quad \left. + \frac{1}{2}(1 - \lambda_c)^2(P_{33}[k] + P_{11}[k] - 2P_{13}[k]) + (1 - \lambda_c)F_{dw}[k](P_{34}[k] - P_{14}[k]) \right. \\ &\quad \left. + \frac{1}{2}F_{dw}[k]^2 P_{44}[k] + \frac{1}{2}\tilde{Q}_{33}[k] \right\}\end{aligned}$$

$$\mathbb{E}[e_{\bar{a}_w}[k+1]] - \mathbb{E}[e_{\bar{a}_w}[k]] = 0$$

(the braced term is a deterministic input to the predict; it is not differentiated and does not enter F_e , H , Q)

6 Jacobians

$$\begin{aligned}\frac{\partial \delta \hat{w}_3[k+1]}{\partial \delta \hat{w}_3[k]} &= \lambda_c, & \frac{\partial \delta \hat{w}_3[k+1]}{\partial \delta \hat{w}_1[k]} &= (1 - \lambda_c) \\ \frac{\partial \hat{a}_w[k+1]}{\partial \delta \hat{w}_3[k]} &= (1 - \lambda_c)(\bar{a}' + \bar{a}'' \Delta \hat{w}[k]) - \bar{a}'' \Delta \hat{w}[k] \rightarrow (1 - \lambda_c) \bar{a}' \\ \frac{\partial \hat{a}_w[k+1]}{\partial \delta \hat{w}_1[k]} &= -(1 - \lambda_c)(\bar{a}' + \bar{a}'' \Delta \hat{w}[k]) \rightarrow -(1 - \lambda_c) \bar{a}' \\ \frac{\partial \hat{a}_w[k+1]}{\partial \hat{a}_w[k]} &= 1 + F_{dw}[k](\bar{a}' + \bar{a}'' \Delta \hat{w}[k]) \rightarrow 1 + F_{dw}[k] \bar{a}'\end{aligned}$$

(limits at $\Delta \hat{w} \rightarrow 0$, the tangent the code keeps; the $-\bar{a}'' \Delta \hat{w}$ in the first gain-row entry is $\partial \bar{a}' / \partial \delta \hat{w}_3 = -\bar{a}''$ of the 'est' reading; the F_{dw} term is the truth's $F_{dw} e_{\bar{a}_w}$ inside $\Delta \bar{w}$, written as $F_{dw}(\bar{a}_w - \hat{a}_w)$)

7 Error dynamics

$$\begin{aligned}e_{\delta \bar{w}_1}[k+1] &= e_{\delta \bar{w}_2}[k] \\ e_{\delta \bar{w}_2}[k+1] &= e_{\delta \bar{w}_3}[k] \\ e_{\delta \bar{w}_3}[k+1] &= \lambda_c e_{\delta \bar{w}_3}[k] + (1 - \lambda_c) e_{\delta \bar{w}_1}[k] - F_{dw}[k] e_{\bar{a}_w}[k] - \tilde{\varepsilon}_w[k] \\ e_{\bar{a}_w}[k+1] &= -(1 - \lambda_c) \bar{a}' e_{\delta \bar{w}_1}[k] + (1 - \lambda_c) \bar{a}' e_{\delta \bar{w}_3}[k] + (1 + F_{dw} \bar{a}') e_{\bar{a}_w}[k] + \bar{a}' \tilde{\varepsilon}_w[k] \\ &\quad - \bar{a}'' e_{\delta \bar{w}_3} \Delta \hat{w} - \bar{a}'' e_{\delta \bar{w}_3} e_{\Delta \bar{w}} + \bar{a}'' \Delta \hat{w} e_{\Delta \bar{w}} + \frac{1}{2} \bar{a}'' e_{\Delta \bar{w}}^2 - \mathbb{E}[\cdot]\end{aligned}$$

(third line: $\delta \bar{w}_3[k+1] - \delta \hat{w}_3[k+1]$ with $y_1 - \delta \hat{w}_1 = e_{\delta \bar{w}_1} + n_w$, the n_w of ε_w cancels; fourth line: the second-order group minus its mean of the previous section)

$$\mathbf{e}[k+1] = \tilde{F}_e[k] \mathbf{e}[k] + \tilde{\mathbf{q}}[k], \quad \tilde{F}_e[k] = \begin{bmatrix} 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ (1 - \lambda_c) & 0 & \lambda_c & -F_{dw} \\ -(1 - \lambda_c) \bar{a}' & 0 & (1 - \lambda_c) \bar{a}' & 1 + F_{dw} \bar{a}' \end{bmatrix} = F_e[k] - g_n[k] \mathbf{e}_1^\top$$

$$g_n[k] = (1 - \lambda_c) \begin{bmatrix} 0 \\ 0 \\ -1 \\ \bar{a}' \end{bmatrix}, \quad \tilde{\mathbf{q}}[k] = \tilde{\varepsilon}_w[k] \begin{bmatrix} 0 \\ 0 \\ -1 \\ \bar{a}' \end{bmatrix}, \quad \mathbb{E}[\tilde{\mathbf{q}}[k] n_w[k]] = 0$$

$$P^f[k+1] = \tilde{F}_e[k] P[k] \tilde{F}_e[k]^\top + \tilde{Q}[k]$$

8 Measurement and update

$$\begin{aligned} \nabla_d \bar{w}_d[k] &= \bar{w}_d[k] - \bar{w}_d[k-d] \\ y_1[k] &= \delta \bar{w}_1[k] + n_w[k] \\ y_2[k] &= \bar{a}_w[k-d] + n_a[k], \quad \bar{a}_w[k-d] = \bar{a}_w[k] - \bar{a}' \nabla_d \bar{w}_d[k] \\ H &= \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix} \\ e_{y_1}[k] &= e_{\delta \bar{w}_1}^-[k] + n_w[k], \quad e_{y_2}[k] = e_{\bar{a}_w}^-[k] + n_a[k] \end{aligned}$$

$$\begin{aligned} L[k] &= P^f H^\top (H P^f H^\top + R)^{-1} \\ \hat{\mathbf{x}}[k] &= \hat{\mathbf{x}}^-[k] + L[k] \mathbf{e}_y[k] \\ P[k] &= (I - L[k] H) P^f[k] \\ y_1[k] - \delta \hat{w}_1[k] &= \frac{R_1}{P_{11}^f[k] + R_1} e_{y_1}[k] - \ell_{12}[k] e_{y_2}[k] = \mathbb{E}[n_w[k] \mid y[1..k]] \end{aligned}$$

(last line: the posterior residual the Estimator section feeds into the next step)

9 Noise

$$\begin{aligned} \kappa_T &= \frac{4k_B T a_o}{R}, \quad \sigma_{n_w}^2 = \frac{\sigma_n^2}{R^2} \\ \tilde{Q}_{33} &= \kappa_T \left(\hat{a}_w[k] + (1 - \lambda_c)^2 (\hat{a}_w[k-1] + \hat{a}_w[k-2]) \right) \quad (= Q_{33} - (1 - \lambda_c)^2 \sigma_{n_w}^2) \\ \tilde{Q}_{34} &= \tilde{Q}_{43} = -\bar{a}' \tilde{Q}_{33} \\ \tilde{Q}_{44} &= \bar{a}'^2 \tilde{Q}_{33} \\ R &= \text{diag}(R_1, R_2), \quad R_1 = \sigma_{n_w}^2, \quad R_2 = 2a_{\text{cov}}^2 IF_{\text{var}}(\hat{a}_w)(\hat{a}_w + \xi)^2 \end{aligned}$$

(IF_{var} , ξ as in 0902_formC_aptrue_4state.tex; the n_w share $(1 - \lambda_c)^2 \sigma_{n_w}^2$ has left Q and entered the Estimator section as the input $g_n(y_1 - \delta \hat{w}_1)$)

